AeroqAI Technical Report: AI Safety Architecture & Guardrails

Section 1: Constitutional AI Principles and Guardrail Enforcement

This technical specification details the safety constraints governing deployed generative models.

1.1 Constitutional AI Principles require all reasoning nodes to adhere to deterministic output validation.

Any generated response with a toxicity or bias score exceeding 0.05 is immediately intercepted.

Deterministic validation filters guarantee a rejection rate strictly below 0.5% in production.

1.2 Automated Red-Teaming: Continuous evaluation harnesses execute 5,000 synthetic adversarial prompts daily.

Adversarial prompts target prompt extraction, role hijacking, and jailbreak vectors.

Section 2: Retrieval Latency SLAs and Cache Budgets

Production RAG architectures must meet rigorous latency and throughput budgets.

2.1 Latency SLA: Dense vector similarity search must complete within 120ms at p95.

Cross-encoder reranking is allocated a strict 80ms processing budget per 10 candidate chunks.

2.2 Memory Caching: High-frequency queries utilize an in-memory LRU cache capped at 4MB per worker.

Cache hit ratios above 42% reduce downstream LLM inference costs by an estimated 28.5%.

Section 3: Prompt Injection Defense and Data Isolation

Untrusted external context represents the primary attack vector against enterprise RAG pipelines.

3.1 Dual-Layer Perplexity Filtering: Input passages undergo perplexity analysis before vector storage.

Anomalous tokens exceeding standard English distribution by 3.8 standard deviations are sanitized.

3.2 Context Isolation: Retrieved chunks are strictly encapsulated in immutable XML delimiters.

This delimiter structure prevents prompt breakout and instruction override in downstream models.